Exact nonlinear slow-mode energy and fixed-amplitude adiabatic metrics

24 September 2026. This note replaces a linearized fiber deformation by an actual horizontal curve of hyperbolic metrics. It determines every Ricci block, the complete nonlinear slow equation, and its normal feedback. The resulting energy identity remains valid at fixed, nonzero moduli amplitude. On cyclic covers it produces pointwise and unnormalized integral Ricci errors tending to zero while that amplitude does not shrink.

The full-mode theorem in I33 concerns arbitrary tensor perturbations in a shrinking neighborhood and measures distance to **all** normalized Einstein metrics. The present theorem treats a larger nonlinear, geometrically specified family, but its fixed-distance conclusion concerns fiber-preserving product outputs. Neither conclusion replaces the other. The underlying smooth products already carry exact Einstein metrics.

The submersion identities and the exact adapted rigidity statement were already available in Round 20. They are not counted as new results. The additions here are the exact arclength equation, the full residual-energy identity including normal acceleration, and its fixed-amplitude application. The companion actual-gauge identity places these metrics inside the first line's actual bilateral monopole near-saturation regime.

1. An actual, horizontally marked hyperbolic family

Let F be a closed oriented surface of genus at least two. Fix a smooth curve h_v, defined on an open interval containing a compact interval J, of metrics of Gaussian curvature -1. Require

$$k_v = \partial_v h_v, \qquad \operatorname{tr}_{h_v} k_v = 0, \qquad \operatorname{div}_{h_v} k_v = 0, \qquad \int_F |k_v|^2_{h_v}\, dA_{h_v} = 1. \tag{1}$$

Such nonconstant curves exist. A regular curve in Teichmüller space has a horizontal lift; after reparametrization by its real-tensor L^2 arclength it satisfies (1). Rupflin–Topping, Definition 1.1 and the discussion of equation (1.1) in *Horizontal curves of hyperbolic metrics*, give the horizontal decomposition. Horizontality is equivalently the real part of a holomorphic quadratic differential. Its length is a fixed normalization of Weil–Petersson length; no conventional WP constant is silently inserted into (1).

Pointwise trace-freeness implies that all h_v have the **same area form**, not just the same total area. Write this form as dA_F, and set

$$V_F = \int_F dA_F = 4\pi(\operatorname{genus}(F) - 1), \qquad N_v = \partial_v k_v - k_v h_v^{-1} k_v,$$

$$C_4(v) = \int_F |k_v|^4_{h_v} dA_F, \qquad C_N(v) = \int_F |N_v|^2_{h_v} dA_F, \qquad \Xi(v) = \frac{C_4(v)}{16} + \frac{C_N(v)}{4}. \tag{2}$$

Here a juxtaposition such as $k_v h_v^{-1} k_v$ denotes contraction as bilinear forms. All these are tensors and functions of the actual marked metrics on F, not only formal Teichmüller coordinates. Differentiating the trace and unit-speed conditions gives

$$\operatorname{tr}_{h_v} N_v = 0, \qquad \int_F \langle N_v, k_v\rangle_{h_v} dA_F = 0, \qquad \Xi(v) \geqslant \frac{1}{16 V_F}. \tag{3}$$

For example, with $A = h_v^{-1} k_v$, differentiation of the squared L^2 norm yields $0 = 2\int_F (\langle \partial_v k_v, k_v\rangle - \operatorname{tr} A^3)$. This proves the middle identity. The last follows from Cauchy–Schwarz and $\int_F |k_v|^2 = 1$. These

identities do not assume that the curve is a WP geodesic. In particular, its normal acceleration is not discarded.

Let (B,γ) be any closed connected oriented Riemannian surface, and let $v : B \to J$ be smooth. On the original smooth product define

$$G = \gamma + h_{v(x)}, \qquad E = \mathrm{Ric}_G + G. \tag{4}$$

The marking in (1) defines a flat horizontal distribution. Throughout, $\Delta_\gamma = -\operatorname{div}_\gamma \nabla$ is nonnegative, and tensor norms without an indicated average are unnormalized norms.

2. Complete nonlinear equation and residual energy

Theorem A. For (1)–(4), all Ricci-error blocks are exactly

$$\begin{aligned} E_{HH} &= (K_\gamma + 1)\gamma - \tfrac{1}{4}|k_v(y)|^2_{h_v}\, dv \otimes dv, \\ E_{HV} &= 0, \\ E_{VV} &= \tfrac{1}{2}k_v \Delta_\gamma v - \tfrac{1}{2}N_v |dv|^2_\gamma. \end{aligned} \tag{5}$$

Consequently the retained slow equation and its entire transverse component along this family are

$$\int_F \langle E_{VV}, k_v\rangle dA_F = \tfrac{1}{2}\Delta_\gamma v, \qquad E_{VV} - k_v \int_F \langle E_{VV}, k_v\rangle dA_F = -\tfrac{1}{2}N_v|dv|^2. \tag{6}$$

They give the exact global identity

$$\boxed{\begin{aligned} \int_{B\times F} |E|^2_G dV_G = \int_B \Big[&\frac{1}{4}(\Delta_\gamma v)^2 + 2V_F \left(K_\gamma + 1 - \frac{|dv|^2}{8V_F}\right)^2 \\ &+ \left(\Xi(v) - \frac{1}{32V_F}\right)|dv|^4 \Big] dA_\gamma. \end{aligned}} \tag{7}$$

In particular, for an arbitrary base metric,

$$\int |E|^2 \geqslant \frac{1}{4}\int_B (\Delta_\gamma v)^2 dA_\gamma + \frac{1}{32V_F}\int_B |dv|^4 dA_\gamma. \tag{8}$$

If $K_\gamma = -1$, the simpler identity is

$$\boxed{\int |E|^2 = \frac{1}{4}\int_B (\Delta_\gamma v)^2 dA_\gamma + \int_B \Xi(v)|dv|^4 dA_\gamma.} \tag{9}$$

There is no potential term depending only on v, and no uncomputed $O(v^2)$ error. The exact family of hyperbolic fibers is responsible for this cancellation.

Proof. In base coordinates x^i and fiber coordinates y^a, put $S_i = \frac{1}{2}h^{-1}\partial_i h = \frac{1}{2}h^{-1}k_v \partial_i v$. Its trace is zero. The mixed Christoffel symbols of the block metric are

$$\Gamma^a_{ib} = \tfrac{1}{2}(h^{-1}k_v)^a{}_b \partial_i v, \qquad \Gamma^i_{ab} = -\tfrac{1}{2}\gamma^{ij}(k_v)_{ab}\partial_j v; \tag{10}$$

the remaining nonzero symbols within each factor are its intrinsic Christoffel symbols. Substitution into Ricci, or the full flat-connection identities in Round 20, gives

$$(\mathrm{Ric}_G)_{ij} = K_\gamma \gamma_{ij} - \mathrm{tr}(S_i S_j), \qquad (\mathrm{Ric}_G)_{ia} = (\operatorname{div}_{h_v} S_i)_a,$$

$$(E_{VV})^\sharp = -\gamma^{ij} D_i S_j, \quad D_i S_j = \partial_i S_j + [S_i, S_j] - \Gamma^\ell_{ij}(\gamma) S_\ell. \tag{11}$$

The mixed term vanishes by (1). The commutator vanishes because every S_i is a scalar multiple of the same endomorphism. Differentiating $h^{-1}k_v$ gives $h^{-1}N_v$. Since $\gamma^{ij}(\partial_i\partial_j v - \Gamma^\ell_{ij}\partial_\ell v) = -\Delta_\gamma v$, (11) proves all of (5).

Equation (3) now proves (6) and

$$\int_F |E_{VV}|^2 dA_F = \tfrac{1}{4}(\Delta_\gamma v)^2 + \tfrac{1}{4} C_N(v)|dv|^4. \tag{12}$$

Set $c = K_\gamma + 1$, $X = |dv|^2$. Since $|dv \otimes dv|^2 = X^2$ and $\langle \gamma, dv \otimes dv\rangle = X$, the horizontal norm integrated over a fiber is

$$2V_F c^2 - \tfrac{1}{2} cX + \tfrac{1}{16} C_4(v) X^2. \tag{13}$$

Completing this scalar square and using (12) proves (7). Equations (8)–(9) follow from (3) and $c = 0$, respectively. □

Corollary A1 (forced slow energy and exact classification). Define $f(x) = \int_F \langle E_{VV}, k_v\rangle dA_F$. Then, for any constant c,

$$\Delta_\gamma v = 2f, \qquad \int_B |dv|^2 dA_\gamma = 2\int_B (v - c) f \, dA_\gamma,$$

$$\|v - \bar v\|_{L^2(B)} \leqslant \frac{2}{\lambda_1(B,\gamma)} \|E\|_{L^2(B\times F)}. \tag{14}$$

Here $\lambda_1 > 0$ is the first nonzero scalar eigenvalue, and the last inequality follows from $\|f\|_2 \leqslant \|E_{VV}\|_2$. No scalar spectral gap has been silently made uniform in the base.

If G is Einstein with **any** constant λ, then $\mathrm{tr}_{VV} \mathrm{Ric}_G = -2$ by (3),(5), so $\lambda = -1$. Equivalently, pairing the Einstein equation with k_v in (6) already implies $\Delta v = 0$. Connectedness of the closed base gives v constant, and then (5) forces $K_\gamma = -1$. Thus the exact solutions in this entire block family are precisely the products. This recovers the relevant inherited adapted rigidity, while (6)–(14) supply its quantitative nonlinear equation and full residual energy.

Corollary A2 (local stress and global amplitude). Put $\delta = \|G^{-1}E\|_{\infty,\mathrm{op}}$. For an arbitrary base,

$$|dv|^2 \leqslant 8V_F\delta, \qquad \mathrm{osc}_B\, v \leqslant \sqrt{8V_F\delta}\ \mathrm{diam}(B,\gamma). \tag{15}$$

When $K_\gamma = -1$, 8 can be replaced by 4. Indeed, the trace-free part of the horizontal block has operator norm $|k_v|^2|dv|^2/8$; its norm is at most δ in dimension two. Taking a fiber maximum and using $\int_F |k_v|^2 = 1$ proves the first bound. For a hyperbolic base the whole horizontal error has norm $|k_v|^2|dv|^2/4$. Integrating along a minimizing base geodesic proves the amplitude estimate. Base-curvature adjustment therefore cannot remove the rank-one trace-free stress.

3. Fixed-amplitude cyclic-cover construction

Fix a closed hyperbolic surface (B, b) and a primitive class in $H^1(B;\mathbb{Z})$. Let η be its harmonic representative, normalized to have integral periods. On the associated infinite cyclic cover B_∞, write $dz = \eta$, with $z(\tau x) = z(x) + 1$ for a deck generator τ. The function z is harmonic. Let $B_m = B_\infty/\langle\tau^m\rangle$, with the lifted hyperbolic metric b_m.

Fix $a > 0$ with $[-a, a] \subset J$, and suppose the two marked hyperbolic structures $[h_{-a}]$ and $[h_a]$ are different. This is achievable on an arbitrarily short regular horizontal arc. Define the actual metrics

$$v_m = a\cos(2\pi z/m), \qquad M_m = B_m \times F, \qquad G_m = b_m + h_{v_m}. \tag{16}$$

The amplitude a is fixed independently of m.

Theorem B. This sequence has uniform local smooth bounded geometry and is noncollapsed on a fixed local scale. For every fixed integer $j \geqslant 0$ and $0 < \alpha < 1$, in the uniformly lifted product charts,

$$\|\operatorname{Ric}_{G_m} + G_m\|_{C^{j,\alpha}_{\mathrm{uloc}}} \leqslant C_{a,j,\alpha} m^{-2}, \qquad c_a m^{-2} \leqslant \delta_-(G_m) \leqslant C_a m^{-2}, \tag{17}$$

$$c_a m^{-3} \leqslant \int_{M_m} |\operatorname{Ric}_{G_m} + G_m|^2 dV_{G_m} \leqslant C_a m^{-3}, \qquad \operatorname{osc}_{B_m} v_m = 2a. \tag{18}$$

The constants depend on the fixed surfaces, harmonic form, and fixed horizontal curve, not on the covering degree. The volume is exactly $m \operatorname{Area}(B, b) V_F$, and diameter is comparable to m. In particular, even unnormalized L^2 Ricci error tends to zero.

Proof. The harmonic chain rule gives exactly

$$dv_m = -a\frac{2\pi}{m}\sin(2\pi z/m)\eta, \qquad \Delta_{b_m} v_m = \left(\frac{2\pi}{m}\right)^2 |\eta|^2 v_m. \tag{19}$$

All derivatives of the lifted η are uniformly bounded. Every fixed number of derivatives of $|dv_m|^2$ and Δv_m is therefore $O(m^{-2})$. The compact smooth hyperbolic path has uniform bounds of every fixed order, so (5) proves the first part of (17). Although the Hessian of v_m need only be $O(m^{-1})$, the complete Ricci formula involves its **trace** in (19); there is no missing $O(m^{-1})$ mixed block, because horizontality makes it vanish exactly.

The metrics G_m are uniformly equivalent to $b_m + h_0$, with all local coefficient derivatives uniformly bounded. Lift a fixed finite atlas on B and a fixed finite atlas on F, shrinking its charts once if necessary. These charts give local ellipticity, curvature bounds and a fixed-radius ball-volume lower bound independent of m. The same argument provides uniform harmonic-coordinate bounds if those are desired, but harmonic coordinates are not needed for (17).

A connected compact fundamental cell for B_∞ has bounded diameter. The cyclic chain of m cells gives $\operatorname{diam}(B_m) \leqslant Cm$. The circle-valued function z modulo m has uniformly bounded differential and is onto the circle of length m, giving the reverse bound. Projection to B_m is distance nonincreasing; a horizontal path at fixed fiber coordinate and then a uniformly bounded fiber path prove the corresponding statements for G_m. Surjectivity also gives $\operatorname{osc} v_m = 2a$. Equation (15) now gives the lower bound in (17).

For completeness the integral rate in (18) is exact in order, not only an upper bound. For $m \geqslant 5$, deck summation gives

$$\sum_{r=0}^{m-1} \cos^2(\theta + 2\pi r/m) = \frac{m}{2}, \quad \sum_{r=0}^{m-1} \sin^4(\theta + 2\pi r/m) = \frac{3m}{8}. \tag{20}$$

Write $I_\eta = \int_B |\eta|^4 dA_b > 0$. It follows that

$$\int_{B_m} (\Delta v_m)^2 dA_{b_m} = \frac{a^2(2\pi)^4}{2m^3} I_\eta, \qquad \int_{B_m} |dv_m|^4 dA_{b_m} = \frac{3a^4(2\pi)^4}{8m^3} I_\eta. \tag{21}$$

Insert these into (9), using the positive lower and finite upper bounds of Ξ on $[-a,a]$, to prove (18). □

This construction is an actual nonlinear continuation of the low channel $q(y)f(x)$; it is not a new smooth-topology supply theorem. It sharpens the earlier shrinking-amplitude test by eliminating the spurious quadratic potential of a linear metric interpolation. Its residual has no fixed amplitude floor because every constant parameter gives an exact Einstein product. All normal acceleration nevertheless remains visible in (9).

4. A precise fixed-distance statement, and its scope

Let $\mathscr{P}_m$ be the metrics on $B_m \times F$ obtained by pulling back a hyperbolic product $\gamma' + h_*$ through any smooth fiber-preserving diffeomorphism $\Phi(x,y) = (\varphi(x), \phi_x(y))$, with orientation-preserving fiber maps. Neither the product parameters nor the diffeomorphisms are fixed. Set

$$d_0(G_m, \mathscr{P}_m) = \inf_{H \in \mathscr{P}_m} \|G_m^{-1}H - I\|_{\infty,\mathrm{op}}, \quad D_T = d_T([h_{-a}],[h_a]) > 0, \tag{22}$$

where $d_T = \frac{1}{2}\inf \log K$ is Teichmüller distance.

Theorem C. For every m,

$$d_0(G_m, \mathscr{P}_m) \geqslant \tanh D_T > 0. \tag{23}$$

Proof. If $\varepsilon = \|G_m^{-1}H - I\|_\infty \geqslant 1$, the claim is automatic. Otherwise $(1-\varepsilon)G_m \leqslant H \leqslant (1+\varepsilon)G_m$. Restricting to a vertical tangent space gives the same comparison between $h_{v_m(x)}$ and $\phi_x^* h_*$, even when H has cross terms. The mapping class of ϕ_x is constant on connected B_m, so all the latter fiber metrics represent a single marked Teichmüller point t_*. The identity on a fiber has dilatation at most $\sqrt{(1+\varepsilon)/(1-\varepsilon)}$, whence

$$d_T([h_{v_m(x)}], t_*) \leqslant \frac{1}{4}\log\frac{1+\varepsilon}{1-\varepsilon}. \tag{24}$$

Use fibers where $v_m = \pm a$ and the triangle inequality. Then $D_T \leqslant \frac{1}{2}\log((1+\varepsilon)/(1-\varepsilon)) = \operatorname{arctanh}\varepsilon$, proving (23). □

Thus no estimate $d_0(G, \mathscr{P}) \leqslant \rho(\|\operatorname{Ric}_G + G\|)$ with $\rho(t) \to 0$ can be uniform over these cyclic covers, even for fixed arbitrarily small but nonzero moduli amplitude, uniform local geometry, and unnormalized L^2 residual tending to zero. The omitted global information is the slow coefficient, not a local curvature spike.

Theorem C does **not** replace $\mathscr{P}_m$ by the set of all Einstein metrics or permit arbitrary non-fiber-preserving diffeomorphisms. Such a replacement would need a theorem showing that every sufficiently close Einstein metric has the required global structure. Round 18 proves local completeness for each fixed background; I33 gives a shrinking uniform neighborhood. Neither supplies that missing fixed-amplitude coverage. The manifolds M_m have the explicit exact output $b_m + h_0$, so these examples do not refute any existence conclusion of Yau26.

5. What has been retained rather than projected away

The scalar equation (6) is exact for the actual geometric family. Its normal term is retained in (5), contributes the nonnegative C_N term in (7), and can be nonzero. Solving $\Delta v = 0$ alone is not asserted to solve an arbitrary full tensor perturbation problem. Within this family, it makes v constant and then all the written vertical and mixed equations vanish; the remaining base equation is precisely $K_\gamma = -1$.

This supplies a complete nonlinear energy for a genuine slow family and a sharp global-size obstruction to uniform adapted repair. It is compatible with the first-line obstruction/existence distinction: local smallness and actual near-saturation may coexist with macroscopic moduli oscillation, although the same smooth topology has an exact output. The companion gauge calculation identifies the actual harmonic classes and their curvature deficits; those are additional verified information, not a presumption that every slow-moduli problem is controlled by them.

6. Reproducible finite checks

`check_nonlinear_slow_moduli.py` recomputes a four-dimensional Ricci matrix from Christoffel symbols for a two-base-variable area-preserving family, checks the normal-acceleration term, arclength pairing and completed-square coefficients, verifies the harmonic-coordinate cancellation and the cyclic averages in (20), and includes negative controls for dropping acceleration or replacing the scalar Laplacian by a Hessian norm. The local coordinate test is not a closed hyperbolic family; the global family is supplied by the horizontal-lift argument above. These finite checks do not certify the Teichmüller theory or any unrestricted Einstein classification.